Uncovering biological relationships through morphological profiling: The JUMP-Cell Painting dataset and tools

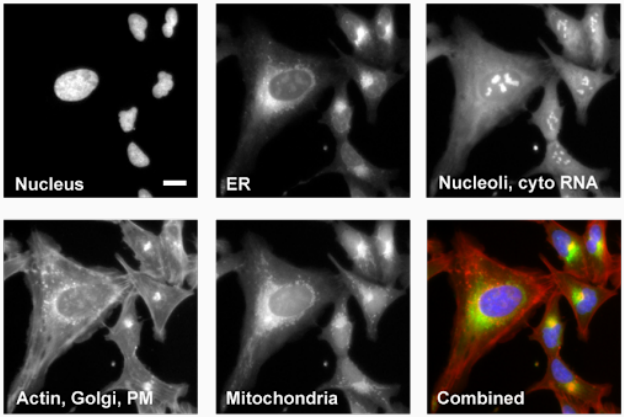
Alán F. Muñoz

Carpenter-Shen Lab, Purdue University; Carpenter-Singh Lab, Broad Institute

10 November 2026

What is Cell Painting?

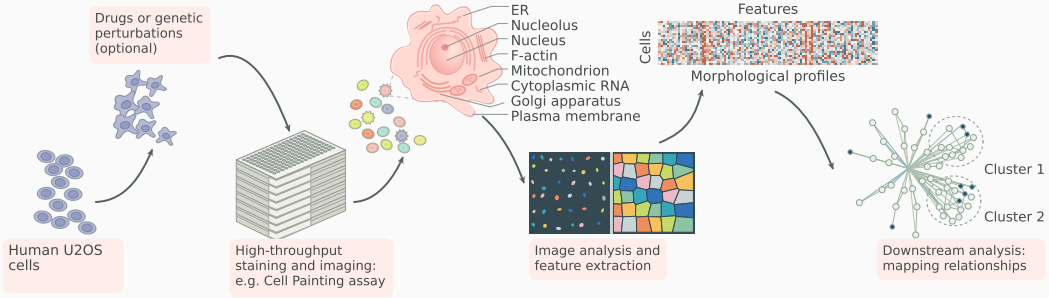
- Fluorescence microscopy of cell morphology.
- Six dyes, five imaging channels.
- Shape, intensity and texture reveal cellular responses.



Source: Muñoz, I2K (2024); assay: Bray et al. (2016)

Cell Painting turns images into measurements

Profiles are numerical descriptions of cells.



Adapted from Chandrasekaran et al. (2021), Fig. 1a

Morphological profiling: How to turn images into profiles

Start with a simple question: are control and perturbed cells different?

Classical features (e.g., CellProfiler)

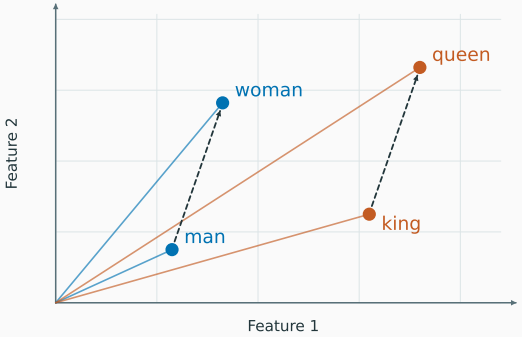
- Size, shape, intensity, texture and relationships between channels and objects.
- Interpretable but partially redundant.

Learned representations

- Features learned from images by neural networks (deep learning).
- Non-redundant numerical vectors (embeddings).

Words as vectors: word2vec

Learning can be viewed as compression



Word → vector

Learned from context

man → woman
king → queen

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$$

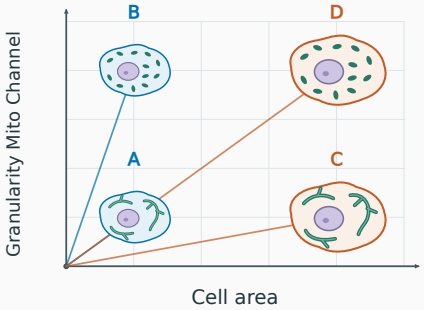
Mikolov et al. (2013)

From words to cells: vectors for comparison

Classical features

Defined measurements

x = [cell area, granularity, ...]

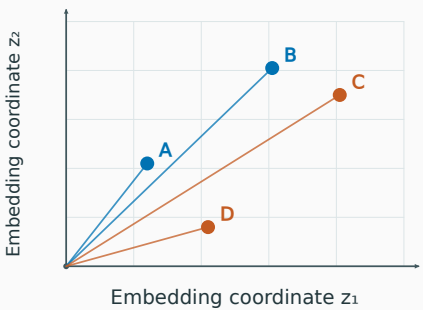


Learned embeddings

Learned from images

Not necessarily directly interpretable.

z = [z_1, z_2, ..., z_n]



Control

Perturbation

JUMP: Joint Undertaking for Morphological Profiling

- Public Cell Painting data: 12 laboratories, around 136,000 perturbations.
- 1.8 billion cells; ~58.3 million images.

CRISPR knockout	Gene overexpression (ORF)	Compound treatment
7,975 gene targets	**12,609** genes	**115,794** compounds

We make JUMP accessible via the JUMP Hub

Choose Your Own Path

1. Just exploring the data?
   - Use a web interface; no programming required. You can look up images for a gene or compound, distinguishing features, most-similar genes or compounds, and more! See the Interactive Tools page for instructions for these tools.
2. Want to fetch and analyze data?
   - Follow our how-to guides for common analyses
3. Looking to build tools?
   - Use our Python libraries cp_measure, cpgdata, jump-portrait, jump-babel and other libraries in the monorepo
   - Access the metadata schema

Biologists can find out:

- Does my gene or compound have a phenotype?
- Which perturbations cause similar phenotypes?
- What changes in cells and organelles?
- Is the phenotype heterogeneous across cells/samples?

Source: JUMP Hub (2026)

Tutorials, how-tos and resources

The Hub contains practical guides to help biologists use JUMP.

How-to guides	No-code gene/compound queries; explore similarity heatmaps.
Tutorials	Curated runnable notebooks for profiles, metadata and statistical analysis.
Further resources	Dataset explanations, protein-localisation analysis and a Guide to Cell Painting for Biologists.

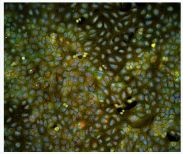
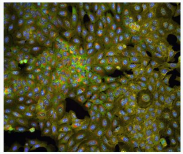
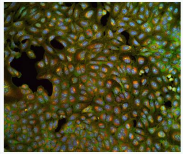
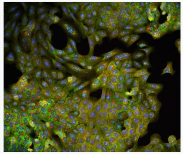
Was my gene or compound tested?

- Find your compound and see whether it was tested.
- Find your gene and see whether it was knocked out or overexpressed.
- View the available profiles, images and metadata.

Link	rowid ▼	JCP2022	NCBI_Gene_ID	standard_key	broad_sample	pert_type
1	1	JCP2022_900002	9	NAT1	ccsbBroad304_00001	trt
2	2	JCP2022_900003	15	AANAT	ccsbBroad304_00002	trt
3	3	JCP2022_900004	18	ABAT	ccsbBroad304_00003	trt
4	4	JCP2022_900005	37	ACADVL	ccsbBroad304_00007	trt
5	5	JCP2022_900006	41	ASIC1	BRDN0001485349	trt
6	6	JCP2022_900006	41	ASIC1	BRDN0001480309	trt
7	7	JCP2022_900006	41	ASIC1	ccsbBroad304_00008	trt

Which perturbations produce related phenotypes?

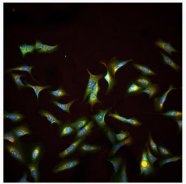
MYT1 example

Gene/Compound	Match	Gene/Compound Example	Match Example	Similarity ▲	Match resources	JCP2022
MYT1	SOS1			0.7195221781730652	External	JCP2022_804400
MYT1	IL13RA2			0.6855948567390442	External	JCP2022_804400

- Precomputed comparisons between perturbation profiles.
- Look for perturbations that match *and* oppose your perturbation of interest.

How does cell morphology change?

AVPR2 overexpression example

Mask	Feature	Channel	Statistic	Gene/Compound	Metadata_image	Median	JCP2022
Cells	Correlation_K_Mito	AGP	8.349928508374835e-29	AVPR2		-0.21309228714749423	JCP2022_905491

- Feature to investigate: mitochondria–AGP overlap.
- Check images: mitochondria versus actin, Golgi and plasma membrane.

Use JUMP as a screen for your phenotype of interest

Feature families

Size / shape	Nuclear area, elongation
Intensity	Mitochondrial staining
Texture	Fine/coarse patterns
Correlation	Between-channel patterns
Distribution	Peripheral/perinuclear signal

Which perturbations change nuclear size?

- Select nuclear area.
- Rank perturbations by feature value.
- Check activity, images and controls.

What do the perturbed cells look like?

MYT1 and PROZ gallery examples

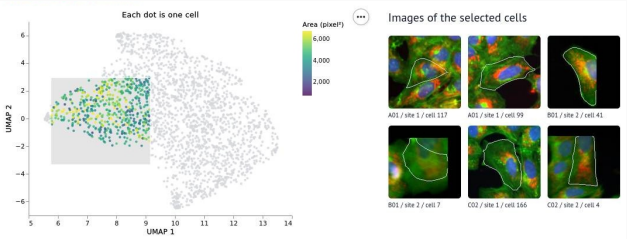
Gene/Compound	External resources	Metadata_JCP2022	Foci 0	Foci 1	Foci 2	Foci 3	Foci 4
PROZ	External	JCP2022_805564					
MYT1	External	JCP2022_804400					
PROZ	External	JCP2022_805564					

Modern tools make image analysis more accessible

Find cells, nuclei and spots	Cellpose (cells and nuclei); Spotiflow (spots)
Follow cells and organelles over time	Trackastra / Ultrack
Measure cell and organelle appearance	cp_measure
Explore your results	marimo

Identify and explore cell subpopulations

Interactive marimo notebook



Each dot is a cell: 2,223 from one Target-2 plate.
Select cells; inspect images, area and other features.

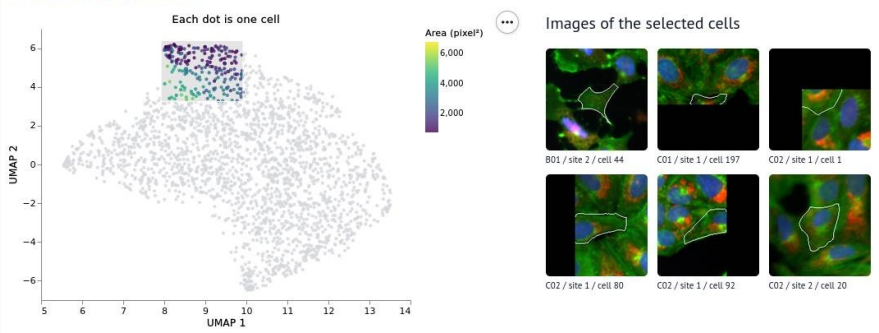
AI can assemble these tools into workflows.

Example prompt: Using the JUMP Hub docs, create a marimo dashboard to link selected cells to images and measurements.

Standard tools support reproducible analysis; biological expertise remains essential.

Source: Muñoz (2026), JUMP Target-2 notebook recording

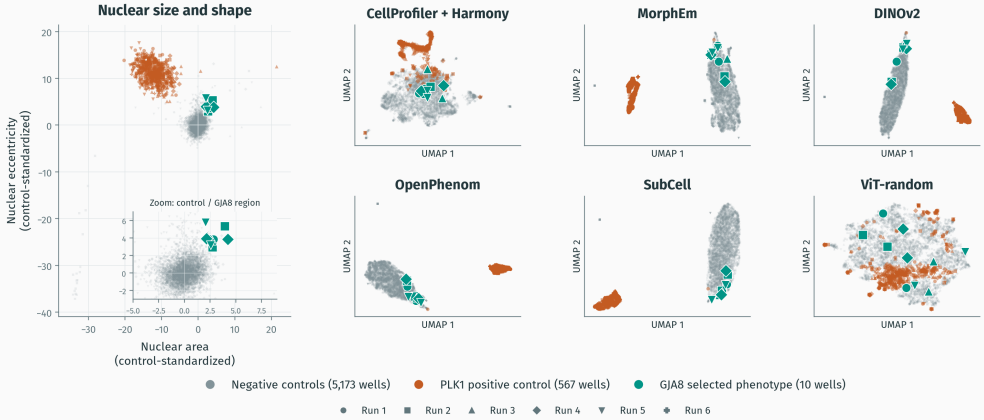
How do cells in another region compare?



- Second selection: 224 cells.
- Compare images and measurements.

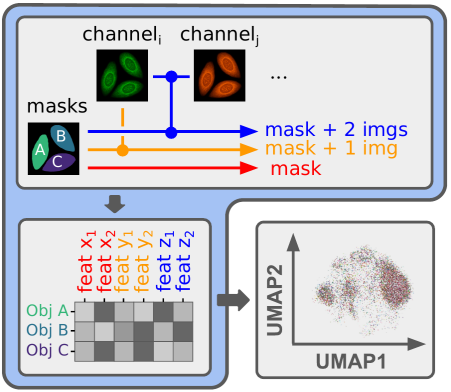
Source: Muñoz (2026), JUMP Target-2 notebook recording

Positive controls contextualize perturbations in both learned and classical representations



Data: JUMP-lite (2026); models: MorphEm, DINOv2, OpenPhenom, SubCell; CellProfiler/Harmony recipe

Not just JUMP: Extract morphological profiles from your own images



cp_measure: feature extraction in Python

- Thousands of features across channels and compartments.
- Fits your Python pipeline.
- Fast software to obtain interpretable morphological features (∼5x CellProfiler).

Adapted from Muñoz et al. (2025), Fig. 1

Applications of JUMP and our tools across biological questions

Evaluate new lysosomal fluorescence probes

Kenneth Li, Vendrell Lab, University of Edinburgh

- Lysosomal acidification probes in cardiomyocytes.
- Uptake, localization and morphology over time.

Compare imaging modalities

Minh Doan, GSK Bioimaging Analytics

- Bright-field time series, Cell Painting, cell cycle and viability.
- Which dyes/imaging methods are better for an different Mechanisms of Action

MisLocus: protein mislocalisation

Johan Fredin, Carpenter-Shen Lab, Purdue University

- Which rare genetic mutations change protein location?
- Compare microscopy of reference and variant proteins.

Integrated in spatial omics software

Tim Treis, Theis Lab, Helmholtz-Munich

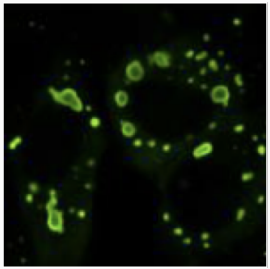
- Integration of morphological features with Squidpy.

MisLocus: a single-cell image benchmark

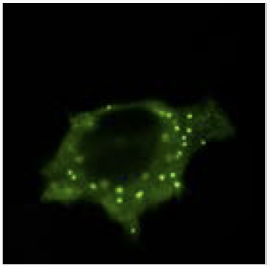
MisLocus contains ∼3 million cells spanning 1,541 reference/variant alleles across 282 genes, excluding controls.

Does a variant change where the protein is located?

LZTR1 reference



LZTR1 G248R



Protein mislocalisation

- Compare reference and variant proteins.
- Test for localisation changes.
- Evaluate image representations.

Unpublished data (preprint coming soon)

Images from Taipale lab

Take-home messages

- JUMP provides public Cell Painting data for around 136,000 chemical and genetic perturbations.
- Biologists can use JUMP Hub to find tested genes and compounds, compare phenotypes and inspect images.
- Open-source image-analysis tools extend morphological profiling to your own experiments.
- Guides and notebooks help scientists or AI agents build analysis workflows.

Thank you

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Carpenter-Shen Lab
Anne E. Carpenter
Runxi Shen

JUMP partners



JUMP supporting partners



JUMP collaborators

Scientific:
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a PerkinElmer Company
Administrative:
Pistoia Alliance
Academic:
Umeå University
Stanford Machine Learning Group

Work on cp_measure and JUMP-lite was supported by GSK and NIH MIRA (R35 GM122547).

References

All named tools and models below are clickable.

Assay, data and resources

- Cell Painting (Bray, 2016); profiling review (2021).
- JUMP consortium; counts; centers; images.
- JUMP-lite paper; data/code.
- JUMP Hub; how-to guides; notebooks.
- Guide to Cell Painting.
- Demo recording; Target-2 artifacts.
- word2vec (Mikolov, 2013).

Image-analysis toolbox

- cp_measure; paper.
- CellProfiler; paper.
- Cellpose; StarDist.
- Spotiflow: spot detection.
- Trackastra; Ultrack.
- Nahual: model interfaces.
- marimo; uv.
- Squidpy; Morpheus.

Representations and methods

- DINOv2; DINOv3.
- MorphEm.
- OpenPhenom.
- SubCell.
- ViT-random: DINOv2 code.
- Harmony; JUMP recipe.
- Parquet; Zarr; JPEG XL.

JUMP Hub and slides

JUMP Hub



broad.io/jump

Slides (PDF)



JUMP-lite: a compact benchmark

Same images. Different methods.



Test whether profiles recover known relationships.



Same selected images: 8.7 TB $\rightarrow$ 92 GB ($\sim$94$\times$ smaller; lossy JPEG XL).

Adapted from Muñoz et al. (2026)

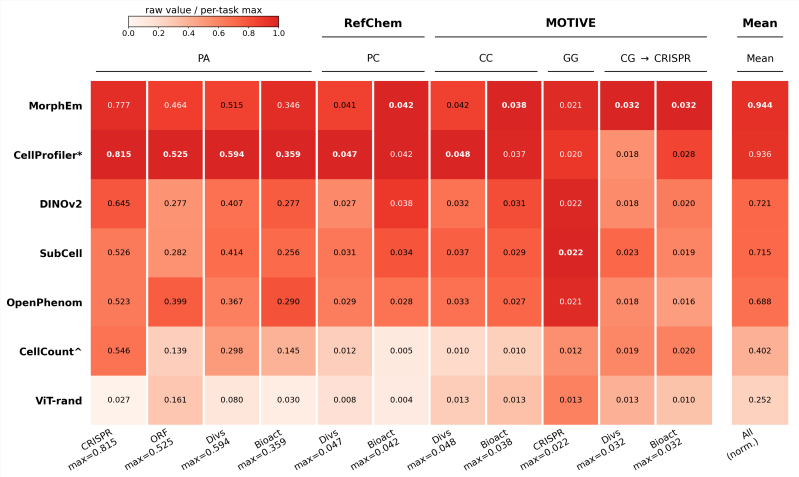
Nahual: swap models within one workflow

Swap models. Keep your workflow.

Image features	DINOv2 ↔ DINOv3 ↔ OpenPhenom ↔ MorphEm
Segmentation	Cellpose ↔ StarDist
Spot detection	Spotiflow
Tracking	Trackastra ↔ Ultrack

JUMP-lite: comparing image representations

Recover replicate perturbations and known biological relationships.



Raw images; colors normalized within each task. CellProfiler/count references use more sites per well.

Adapted from Muñoz et al. (2026), Figure 5